Developing Discipline-Related Teaching Strategies

Competency Information

Competency Description

Teaching a student's specific discipline poses numerous challenges for instructors. The primary challenge is that no universal teaching strategy can be employed to impart learning material to all students effectively. Instead, instructors must utilize multiple teaching strategies that cater to different learning styles. Most of these teaching strategies in the STEM discipline implement active learning approaches (i.e., Modeling instruction, Problem-Based Learning, Argument-Driven Inquiry) designed to meet specific learning objectives. However, instructors must be careful when implementing these active learning approaches. Often instructors focus on the development of fun/enjoyable active-learning activities, and neglect how these activities will lead to any understanding in our students. Effective teaching strategies should promote student understanding by providing a clear purposes and explicit performance goals.

Competency Development

To enhance my proficiency in discipline-related teaching strategies, I enrolled in Teaching College Science (ISE870; Spring 2022). During the course, I had multiple opportunities to evaluate and challenge current pedagogical issues in science education. In addition, I was introduced to various theories and practices in postsecondary teaching, including student-centered pedagogical practices, a range of assessment strategies, and curriculum and lesson design fundamentals. Specific to the development of this competency, I engaged in…

1) *Syllabus Design*: I developed a curriculum for an undergraduate plant Physiology course that showcased formative-heavy teaching (i.e., whiteboarding, feedback,), quantitative reasoning exercises using real-world problems in the discipline, and cooperative learning groups.
2) *Lesson Plan Design & Micro-Lecture:* I developed a student-centered lesson for photosynthesis unit, that would be a part of a broader undergraduate plant physiology course. I implemented Backwards Design when constructing this unit to align learning objectives to the performative task.

Additionally, and was selected to be a FAST fellow in 2023-2024. During this year-long fellowship I received pedagogic mentorship, read educational literature, engaged in group discussion with my FAST cohort and faculty with expertise in undergraduate education. Specific to the development of this competency, I engaged in…

1) *Backwards Design*: As a cohort we read select chapters from Understanding by Design (Wiggins, & McTighe, 2005). This book was critical for my developing Discipline-Related Teaching Strategies as it discussed how to design educational units focused on clear purposes and explicit performance goals, while avoiding *weak educational design* & *excess coverage*.

2) *Led a Cooperative Learning Activity*: I was given the opportunity to teach a 25-minute microlesson to my FAST cohort and the FAST/CRITL steering committee with the goal of incorporating/demonstrating cooperative learning in the lesson. I taught a lesson from a photosynthesis module in Plant Physiology I developed in ISE870.
3) *Designed Discipline Specific Content*: As a part of my Teaching-as-Research project, I developed three quantitative reasoning exercises aligned with learning modules (plant-water relations, photosynthesis, and metabolism) in an upper-level Plant Physiology course. Each of these exercises incorporated problem-based learning (i.e., case studies).

Artifact

Artifact 1. FAST Fellow meeting in the Fall of 2023.

Interpretation / Reflection

I developed discipline-related teaching strategies competency by understanding the theories involved in these teaching strategies and demonstrating proficiency in practical experiences with my peers. Much of the knowledge base I developed in ISE870 was based on active learning practices in science education. The teaching strategies covered in the most detail in the course were modeling instruction, problem-based learning, and argument-driven inquiry. Additionally, we learned how to keep students in the zone of proximal development with these strategies and how to modify instruction if students were to fall off. I implemented these practices when designing a syllabus and microteaching lessons I prepared for my peers. During my fellowship, I applied these teaching strategies in my Teaching-as-Research project when developing discipline specific content. These teaching strategies and their alignment to learning objectives are critical in postsecondary education provide students with a clear purposes and explicit performance goals.